



A Monte Carlo simulation study of the type-I-error and power of ANOVA F-test, Friedman test and randomization test for randomized complete block design

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Abstract

In this work, a Monte Carlo simulation study was conducted to evaluate the type-I-error rate and power of ANOVA F-test (F-Test), Friedman test (FR-Test) and randomization test (R-Test) for the analysis of randomized complete block design. The performance of the F-Test, FR-Test and R-Test was investigated using four effect sizes ($\Delta = 0.5, 0.75, 1$ and 1.5), and three treatments conditions ($t = 3, 6$ and 9) that were simultaneously varied with the number of blocks ($b = 5, 10, 15, 20, 30, 40$ and 50) under three variance ratios ($\sigma^2 = 1, 1.5$ and 3) when the error is normal or skewed (Cauchy and logistic distributions). The simulation results showed that the R-Test was robust in controlling type-I-error rates at 1% and 5% levels of significance and was more powerful than both the ANOVA F-test and Friedman test for most conditions. However, under the Cauchy distribution, the R-Test showed slight limitations at $t = 6$, with increased type-I-error and reduced power. Hence, the R-Test is highly recommended for RCBD due to its superior performance across diverse conditions.

Keywords Monte Carlo · Power · Type-I-error · ANOVA · Randomization test · Robustness · Friedman test

1 Introduction

Analysis of variance F-test (ANOVA F-test) is the uniformly most powerful unbiased (UMP) test for hypothesis test of two or more population means when all underlying assumptions of normality, independence and homogeneity of variance are met (Moder et al. 2013; Macunluoğlu and Ocakoğlu 2023). However, serious challenges are often encountered in the use of ANOVA F-test for analyzing data generated by experimental designs when the assumptions of normality and homogeneity of variances are violated (Marcinko

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2014). The use of ANOVA F-test when these assumptions are violated, often lead to passing invalid inference thereby making ANOVA F-test inadequate for analyzing data generated by experimental designs (Mewhort 2005; Önder 2007; Moder 2010; Wijesuriya et al. 2017; Oladugba et al. 2023). Three solutions are generally recommended for the situations in which the assumptions of ANOVA F-test are not met. These are: subjecting the data to an appropriate transformation; using the non-parametric counterparts; and applying resampling methods like permutation test (Mendes and Akkartal 2010). In situations where the distribution of the error is moderately skewed either to the left or to the right; an appropriate transformation of the dataset (using log, square root and arc-sine transformations etc.) enables one to still use the ANOVA F-test (Mahapoonyanont et al. 2010; Ribeiro-Oliveira et al. 2018). Although, the foregoing transformation techniques often enables one to normalize the distribution of the error; it is important to note that data transformation is not suitable for every dataset. Furthermore, difficulty in interpretation and discussion of results on scales other than the original data is one of the problems associated with data transformation. In a case of gross departures from normality, the inability to normalize the distribution of the error after each transformation makes it difficult for the ANOVA F-test to be used in such scenario. In order to overcome this scenario, several non-parametric methods have been recommended to be used for the analysis of the data generated by randomized experimental designs (Fawcett and Salter 1984; Zimmerman and Zumbo 1993; O’Gorman 2001; Moder 2010; Rosa et al. 2012; Xu et al. 2017; Nwobi and Akanno 2021; Macunluoğlu and Ocakoğlu 2023). Non-parametric methods such as Wilcoxon rank-sum test, Kruskal–Wallis and Friedman tests etc., are statistical tests that do not assume that the data follow a specific distribution. This implies that non-parametric methods are robust to the departures of normality since they do not depend on the assumption of normality. However, although not readily acknowledged, non-parametric methods are constrained by the assumption of homogeneity of variance and whenever this assumption is not met, there is generally a loss in the power of non-parametric tests (Moder 2010). Under certain parametric assumption violations, the permutation test has been proposed as an alternative to the ANOVA F-test to rehabilitate its robustness and superior in power to the non-parametric alternative (Mewhort 2005; Önder 2007; Butar 2008; Mendes and Akkartal 2010; Oladugba, et al. 2023).

Permutation test is a distribution-free test used in hypothesis testing in which under the null hypothesis, the p value is the proportion of data permutations yielding a test statistic as large as the observed test statistic. In hypothesis testing, permutation test can either be exact or approximate. A permutation test is said to be exact if all possible permutations of the observed values were enumerated in order to empirically obtain an exact p value and approximate, if a random sample of all possible statistic values were used to obtain its p value (Peres-neto and Olden 2001; Edgington and Onghena 2007). For the purpose of this work, the approximate permutation test also known as randomization test was adopted due to its seldom unrecognized potential in hypothesis testing of two or more population means. Randomization test has several advantages over parametric and non-parametric tests in hypothesis testing of two or more population means. Firstly, randomization test does not make any assumption about the distribution of the data unlike parametric tests. Secondly, randomization test makes use of the original data unlike non-parametric tests. Several works have been done to compare the type-I-error rate and power of randomization test with that of ANOVA F-test and few with non-parametric tests (Mewhort 2005; Mendes and Akkartal 2010; Wijesuriya et al. 2017; Oladugba et al. 2022; Oladugba et al. 2023). The forego-

ing works showed the superiority of the power of randomization test over ANOVA F-test regardless of sample sizes and effect sizes (Mendes and Akkartal 2010; Wijesuriya et al. 2017), under skewed error (Mewhort 2005; Oladugba et al. 2023), and in the presence of an outlier (Oladugba et al. 2022). However, it is important to note that randomization test do not necessarily allow one to relax the assumption of homogeneity of variance in hypothesis testing of two or more population means despite its superiority in terms of power over parametric and non-parametric tests (Mendes and Akkartal 2010; Wijesuriya et al. 2017).

Several studies had compared the ANOVA F-test with several non-parametric alternatives such as Kruskal–Wallis, Friedman test and randomization test in terms of power and type-I-error rate when the distribution of the error, variance, effect and sample sizes were varied. However, none of the aforementioned tests has been verified to be the appropriate statistical method for the analysis of data from randomized complete block design (RCBD) when the error was skewed with unequal variances. Also, since parametric and non-parametric tests are affected by the degree to which the data do not meet the assumptions of normality and homogeneity of variance, and the randomization test, by the degree of variation in the data, it is appropriate to compare the three tests in terms of type-I-error rate and power in order to validate any of them as the most appropriate in any situation. Hence, the purpose of this work is to evaluate via a Monte Carlo simulation the robustness of type-I-error rate and power of ANOVA F-test, Friedman test and randomization test for the analysis of data generated using RCBD when the error is normal/skewed with equal/unequal variances.

2 Methodology

The ANOVA F-test, Friedman test and randomization test used for the analysis of RCBD are discussed in this section.

2.1 ANOVA F-test (F-test)

Randomized complete block design (RCBD) is one of the basic efficient experimental designs that uses one blocking variable to group heterogeneous experimental units into a more homogeneous units in blocks such that each block contains all the treatments. Suppose there are t -treatments, b -blocks and one observation per treatment in each block; the model for testing the equality of the treatment means in a RCBD is given by

$$Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}; i = 1, 2, \dots, t; j = 1, 2, \dots, b$$

$$\sum_{i=1}^t \tau_i = \sum_{j=1}^b \beta_j = 0; \varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2) \quad (1)$$

where, Y_{ij} is the observed response taken under the i^{th} treatment in the j^{th} block; μ is the grand mean; τ_i is the i^{th} treatment effect; β_j is the j^{th} block effect; ε_{ij} is the random error associated with Y_{ij} .

In this work, the classical RCBD, where each treatment appears exactly once in each block, and each observation is assumed to be independent was considered. It is important to clarify that the RCBD setup in this work does not involve repeated measurements within

blocks. In studies where repeated measures are applied within blocks (e.g., in behavioural or longitudinal designs), the assumption of independence among observations within a block is violated, and additional assumptions are needed for the F-distribution to be valid (Kirk 1995, pp. 271–282). However, since this work does not involve repeated observations within blocks, the classical assumption of independence holds, and the standard ANOVA F-test is valid under the simulated design.

The hypotheses of interest are

$$\begin{aligned} H_0 : \tau_1 &= \tau_2 = \dots = \tau_t = 0 \\ H_1 : \tau_i &\neq 0, \text{ for at least one } i. \end{aligned}$$

If the null hypothesis of no difference in treatment means (effects) is true, the ANOVA F-test statistic (F-Test) is defined as

$$F_0 = \frac{MST_{rts}}{MSE} \sim F_{(t-1), (t-1)(b-1)} \quad (2)$$

where $MST_{rts} = \frac{SS_{Trts}}{t-1}$; $MSB = \frac{SS_B}{b-1}$; $MSE = \frac{SS_E}{(t-1)(b-1)}$ and the sum of squares are defined as follows: $SST_{rts} = \sum_{i=1}^t \frac{Y_{i.}^2}{b} - \frac{Y^2}{tb}$; $SSB = \sum_{j=1}^b \frac{Y_{.j}^2}{t} - \frac{Y^2}{tb}$; $SST = \sum_{i=1}^t \sum_{j=1}^b Y_{ij}^2 - \frac{Y^2}{tb}$; $SS_E = SST - SS_{Trts} - SS_B$.

The H_0 is rejected if $F_0 > F_{\alpha, (t-1), (t-1)(b-1)}$ and conclude that there are differences in the treatment means (effects).

2.2 Friedman test (FR-test)

Friedman test (FR-Test) is a widely used rank-based alternative to the ANOVA F-test for identifying treatment differences in RCBD. The Friedman test does not require that the distribution of the observations across the t -groups within a block are normally distributed. In fact, no particular parametric form is assumed for the distribution of the observations in each group. However, Friedman test assumes that the observations in blocks are mutually independent and the samples are randomly selected from a given population (O’Gorman 2001; Laurent and Turk 2013). For a RCBD with b -blocks, t -treatments in each block and one observation per cell; let R_{ij} denote the rank of the response on the i^{th} treatment within block j , where $i = 1, 2, \dots, t$; $j = 1, 2, \dots, b$ and let $R_{i.}$ be sum of the ranks in the i^{th} treatment. Then, the Friedman test statistic is

$$F_{b,t} = \frac{12r}{t(t+1)} \sum_{i=1}^t (\bar{R}_{i.} - \bar{R}_{..})^2 \quad (3)$$

where $\bar{R}_{i.}$ and $\bar{R}_{..}$ are the mean rank of the i^{th} treatment and the overall mean rank of the t -treatments respectively. If ties exist in any block, then the test statistic becomes

$$F_{b,t}^* = \frac{F_{b,t}^*}{\sum_{r=1}^k (t_r^3 - t_r)} \quad (4)$$

$$1 - \frac{r=1}{N(t^2-1)}$$

where, k is the number of groups with tied observations; t_r is the number of tied observations in the i^{th} tied group and N is the total number of observations. If the number of blocks increases infinitely, the distribution of the $F_{b,t}$ approaches χ_{t-1}^2 under the null hypothesis in RCBD. Hence, the null hypothesis is rejected if $F_{b,t} > \chi_{\alpha,t-1}^2$.

2.3 Randomization test (R-test)

Permutation test is distribution-free test that rely on exchangeability of the experimental data in order to construct a reference distribution for the test statistic under the null hypothesis. In hypothesis testing, permutation test can either be exact or approximate. Exact permutation tests are permutation test which enumerate all possible permutations of the observed values in order to empirically obtain an exact p value (Mewhort 2005; Butar 2008). In the context of t -sample problems, the procedure of the exact permutation tests are as follows:

- Suppose there are N -observations among t -treatments and n_i observations in treatment i , $i = 1, 2, \dots, t$.
- calculate the test statistic for the original data set, called it TS_{obs} . All possible permutations are $\frac{N!}{n_1!n_2!\dots n_t!}$;
- calculate the statistic for each permutation.
- obtain the p value as the proportion of the statistic that are greater than or equal to TS_{obs} . This p value is exact since it was calculated from all possible permutations and it is usually referred to as the significance of the test (Butar 2008).

As N increases, the number of permutations can become extremely very large and, consequently, one may decide to take a random sample of all possible statistic values. In such scenario, the permutation test is known as approximate permutation test or randomization test. Randomization test (R-Test) is an approximate permutation test in which the experimental data are randomly allocated to two or more groups in order to test hypotheses about treatment effects in a randomized experiment (Mewhort 2005). The R-Test method for the analysis of RCBD involves the following procedures:

The hypotheses of interest are

$$H_0 : \tau_1 = \tau_2 = \dots = \tau_t = 0 \text{ Vs } H_1 : \tau_i \neq 0, \text{ for at least one } i.$$

$$\alpha = \alpha_0, \text{ where } \alpha_0 = 0.01, 0.05.$$

If there are no treatment effects, then the experimental data can be repeatedly permuted to test the null hypothesis.

The steps are as follows:

- Partition the N observations (experimental data) into t - treatments and b - blocks such that each block contains exactly one observation for each treatment.

- Select a test statistic that sufficiently explains the experimental data. That is, F-Test of the original observations.
- Compute the test statistic (F-Test) for the experimental data. Denote its value as test statistic_{obs}.
- For each block, randomly shuffle the treatment labels and assign the observations to the new treatment labels to create a permuted dataset.
- Compute the F-statistic of the permuted dataset. If this test statistic is greater than test statistic_{obs}, increment counter.
- Repeat the preceding two steps n times (where n is the number of simulations) to generate an empirical distribution of F-statistics under the null hypothesis.
- Compute the proportion of the F-statistic that are greater than or equal to test statistic_{obs}. Denote this value as the p value.
- Reject the null hypothesis (H_0) if the calculated p value is less than the chosen significance level (α) and fail to reject H_0 if otherwise.

3 Monte Carlo simulation

The Monte Carlo simulation was conducted in order to evaluate the type-I-error rate (in terms of robustness) and power of F-Test, FR-Test and R-Test under the following scenarios listed in Table 1. The normal data were simulated from $N(0,1)$ while the skewed data were simulated from Cauchy(0,1) and Logistic(0,1). Four variables were manipulated: (i) distribution of the error (ii) variance (iii) effect size (iv) sample size. The performance of the F-Test, FR-Test and R-Test were investigated using four effect sizes ($\Delta=0.5, 0.75, 1$ and 1.5), and three treatments conditions ($t=3, 6$ and 9) that were simultaneously varied with the number of blocks ($b=5, 10, 15, 20, 30, 40$ and 50) under three variance ratios ($\sigma^2 = 1, 1.5$ and 3) when the error is normal or skewed. The standard normal distribution was used for the scenarios under which normality assumption holds while Cauchy and logistic distributions were used in cases where the distribution of error was skewed.

The R statistical package was used to implement the Monte Carlo technique sampling of 10,000 permutations from the possible $\frac{(tr)!}{(r!)^t}$ permutations for the R-Test. In the simulation, the experiment was repeated 10,000 times for each distribution. In each repetition, the simulated data were analyzed appropriately using the F-Test, FR-Test and R-Test to obtain the type-I-error rate and power.

The comparison procedures were considered in two categories. Firstly, if the null hypothesis ($H_0 : \tau_i = 0$) was true, the rejection rate of the null hypothesis was considered as the type-I-error rate for each test. The test that has the closest type-I-error to the nominal α_0 , where $\alpha_0 = 0.05$ would be considered to be more robust in terms of type-I-error. Secondly, if the alternative hypothesis ($H_1 : \tau_i \neq 0$) was true, the rejection rate of the null hypothesis

Table 1 The 4 scenarios for both normal and skewed distributions

S/N	Type of error	σ^2
1	Normal	Equal
2	Normal	Unequal
3	Skewed	Equal
4	Skewed	Unequal

was considered as the power for each test. The test that has the larger power would be taken to be more powerful than the other.

4 Results

The type-I-error rates and power of F-Test, FR-Test and R-Test using Monte Carlo simulation were evaluated at a nominal $\alpha=5\%$ for $t=3, 6$, and 9 under normal/skewed errors with equal/unequal variances. The results obtained for normal error with equal/unequal variances are presented in Tables 2, 3, 4 and 5 while skewed errors (Cauchy and Logistic) with equal/unequal variances are presented in Tables 6, 7, 8, 9, 10, 11, 12 and 13. In Tables 2, 6, and 10 the values in bold denote the p values that did not exceed the specified level of significance while in Tables 3, 4 and 5, 7, 8 and 9 and 11, 12 and 13, the values in bold denote the p values that exceeded the power value of 80%.

4.1 Normal

It was observed from Table 2 that the type-I-error rates of all the tests had no specific pattern when the number of treatments were varied. Also, for equal variance, R-Test gave more reliable results compared to F-Test and FR-Test respectively. Under unequal variance, R-Test had more control of type-I-error rates than F-Test. Moreover, under equal/unequal variances, the type-I-error rates of R-Test and FR-Test are the same. The results in Tables 3, 4 and 5 showed that the power of all the tests increases as number of treatments and effect size increases. Furthermore, for equal variance, the power of all the tests increases as the effect size increases under each treatment but had no specific pattern under unequal variance as the effect size increases. Regardless of the variance ratios, the power of R-Test was higher than both F-Test and FR-Test respectively. Although, the power of R-Test was approximately the same with that of F-Test and FR-Test under equal variance.

4.2 Cauchy

The results presented in Table 6 showed that the type-I-error rates of all the tests had no specific pattern when the number of treatments were varied. Furthermore, for equal/unequal variances, the type-I-error rates R-Test was more conservative (i.e., having type-I-error rates below the nominal α of 5%) than that of ANOVA F-test and Friedman test. Also, the results in Tables 7, 8 and 9 showed that the power of all the tests had no specific pattern when the number of treatments were varied. Furthermore, for equal/unequal variances most of the power values of all the tests did not exceed the power value of 80%. The power of R-Test was higher than that of F-Test and FR-Test respectively regardless of the variance ratios.

4.3 Logistic

The results in Table 10 showed that the type-I-error rates of all the tests had no specific pattern when the number of treatments were varied. For equal variance, all the tests maintained relevant type-I-error rates except at $t=3$, where the type-I-error rates of F-Test were quite

conservative and gave more reliable results than R-Test. Under unequal variance, R-Test gave more reliable results compared to F-Test and FR-Test respectively. Furthermore, from the results presented in Tables 11, 12 and 13 it was observed that the power of all the tests increases as number of treatments and effect size increases. Also, under equal/unequal variances most of the power values of all the tests exceeded the power value of 80%. Irrespective of the variance ratios, the power of R-Test was higher than that of F-Test and FR-Test respectively.

5 Discussion of results

The evaluation of the most widely used statistical tests for the analysis of data generated by using randomized experimental designs is appropriate in order to know the test(s) to use under certain parametric assumption violations. The ANOVA F-test, Friedman test and randomization test are statistical tests that can be used to test the same hypothesis in RCBD. In order to determine the appropriate test to use for the analysis of the data generated using RCBD, the robustness and power of R-Test, F-Test and FR-Test were evaluated under normal/skewed errors with equal/unequal variances.

For normal error, it was observed that the type-I-error rates of all the tests were within the range of 0.0226 to 0.0664 for a nominal alpha of 5% which implies that all the tests were robust to the violation of constant variance assumption. Furthermore, for equal variance, R-Test was observed to be more robust than F-Test while the FR-Test was approximately as robust as the R-Test. For unequal variance, R-Test was more robust than F-Test and FR-Test respectively. For data sampled from Cauchy and logistic distributions, the type-I-error rates of all the tests were observed to fall within the range of 0.0000–0.0650 for a nominal alpha of 5% which implies that all the tests were robust to the violation of normality assumption. For skewed errors (Cauchy and logistic) under equal/unequal variances, R-Test was more robust than F-Test and FR-Test respectively. Although, under skewed error (logistic) F-Test gave more reliable results than R-Test at $t=3$. It is important to note that based on the simulation results the type-I-error rates of FR-Test and R-Test were similar regardless of the variance ratios and distribution of the error. The foregoing statement suggest that the R-Test can be used replacement of the both F-Test and FR-Test under normal/skewed errors and equal/unequal variances since the type-I-error rates of FR-Test and R-Test are the same. Hence, the difficulty in interpretation and discussion of results on scales other than the original data associated with data transformation or ranking can be solve with the use of R-Test.

For normal error under equal variance, all the tests were equally powerful. Furthermore, all the tests gave more power values that exceed the power value of 80% as the effect sizes increases in each treatment. The R-Test was observed to be more powerful than both F-Test and FR-Test respectively as number of treatments and effect size were varied under unequal variance. Also, when the effect sizes were varied, the power of all the tests slightly decreased and had less power values that exceed the power value of 80% in each treatment. Generally, the power of all the tests decreases as the variance ratio increases for each effect size normal error. For data sampled from Cauchy and logistic distributions the power of all the tests had no specific pattern as the number of treatments and effect size increases. Also,

under equal variance, all the tests were observed to have similar powers while R-Test was observed to be more powerful than both F-Test and FR-Test respectively under unequal variance. Furthermore, regardless of the variance ratios, all the tests had few power values that exceed the power value of 80% under skewed error (Cauchy) while all the tests had more power values that exceed the power value of 80% under skewed error (logistic). Moreover, under skewed error (logistic) for equal/unequal variances the R-Test was more powerful than F-Test and FR-Test respectively as the number of treatments increases. Although, the power of all tests was observed not to exceed the power value of 80% on average under skewed error (Cauchy), but R-Test gave more power values that exceed the power value of 80% than both F-Test and FR-Test respectively as the number of treatments and effect sizes were varied. Hence, R-Test was observed to be more powerful than both F-Test and FR-Test respectively under skewed errors.

6 Conclusion

This work evaluated the type-I-error rate (in terms of robustness) and the power of the randomization test in comparison with the ANOVA F-test and the Friedman test under normal and skewed error distributions, with both equal and unequal variances, in a RCBD (where observations are assumed to be independent and no repeated measurements occur within blocks) using Monte Carlo simulation. The simulation results showed that all the tests were robust under normal and skewed errors. However, the randomization test was found to be more robust and powerful than the ANOVA F-test and the Friedman test under equal and unequal variances regardless of the distribution of the error. These results underscore the value of adopting the randomization test for analyzing RCBD data. Unlike traditional methods, the randomization test remains robust and reliable under diverse experimental conditions, making it particularly suitable for applications in any experimental research. Therefore, based on the simulation results obtained in this work, the randomization test is recommended for analyzing RCBD data under normal or skewed errors and equal or unequal variances.

The work can be extended to explore scenarios where repeated measurements are introduced within blocks, a situation where the independence assumption is violated. This extension would allow for a more comprehensive evaluation of how within-block correlation affects the validity and performance of these statistical tests, especially in the context of RCBD.

Appendix: R code for simulations

The R code used to perform the simulations in this study is available at the following link: https://drive.google.com/drive/folders/1faVAv4Q1k_HuPDdB8WCbKFvBDaJnHDxS?usp=drive_link. This code allows for the replication of the analyses and can be adapted for similar statistical experiments.

See Tables 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 and 13.

Table 2 Type-I-error rate of F-Test, FR-Test and R-Test under normal error

Number of treatments		$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1$			$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1.5$			$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 2 : 3$		
		F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test
$t=3$	b									
	5	0.0503	0.0392	0.0392	0.0542	0.0289	0.0289	0.0664	0.0226	0.0226
	10	0.0511	0.0445	0.0445	0.0540	0.0483	0.0483	0.0654	0.0489	0.0489
	15	0.0458	0.0525	0.0525	0.0579	0.0514	0.0514	0.0636	0.0489	0.0489
	20	0.0526	0.0506	0.0506	0.0565	0.0499	0.0499	0.0649	0.0487	0.0487
	30	0.0540	0.0496	0.0496	0.0590	0.0482	0.0482	0.0698	0.0480	0.0480
$t=6$	40	0.0517	0.0517	0.0517	0.0552	0.0528	0.0528	0.0609	0.0535	0.0535
	50	0.0504	0.0487	0.0487	0.0491	0.0489	0.0489	0.0585	0.0493	0.0493
	5	0.0499	0.0415	0.0415	0.0530	0.0411	0.0411	0.0607	0.0416	0.0416
	10	0.0541	0.0444	0.0444	0.0585	0.0483	0.0483	0.0660	0.0467	0.0467
	15	0.0528	0.0409	0.0409	0.0579	0.0410	0.0410	0.0639	0.0412	0.0412
	20	0.0481	0.0401	0.0401	0.0554	0.0396	0.0396	0.0626	0.0395	0.0395
$t=9$	30	0.0503	0.0411	0.0411	0.0590	0.0424	0.0424	0.0660	0.0417	0.0417
	40	0.0521	0.0447	0.0447	0.0555	0.0461	0.0461	0.0633	0.0450	0.0450
	50	0.0510	0.0401	0.0401	0.0558	0.0415	0.0415	0.0599	0.0419	0.0419
	5	0.0490	0.0477	0.0477	0.0561	0.0485	0.0485	0.0623	0.0470	0.0470
	10	0.0510	0.0446	0.0446	0.0584	0.0454	0.0454	0.0646	0.0447	0.0447
	15	0.0532	0.0418	0.0418	0.0609	0.0428	0.0428	0.0666	0.0429	0.0429
	20	0.0490	0.0414	0.0414	0.0575	0.0420	0.0420	0.0639	0.0431	0.0431
	30	0.0502	0.0455	0.0455	0.0583	0.0459	0.0459	0.0632	0.0471	0.0471
	40	0.0475	0.0470	0.0470	0.0556	0.0460	0.0460	0.0592	0.0496	0.0496
	50	0.0492	0.0426	0.0426	0.0563	0.0447	0.0447	0.0605	0.0432	0.0432

Table 3 Power of F-Test, FR-Test and R-Test under normal error for $t=3$

Effect sizes	$t(b)$	$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1$			$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1.5$			$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 2 : 3$		
		F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test
$\Delta=0.50$	3(5)	0.1766	0.3024	0.6202	0.1485	0.3784	0.5456	0.1571	0.1877	0.5661
	3(10)	0.6858	0.7505	0.9805	0.6181	0.7505	0.9704	0.3858	0.2831	0.8765
	3(15)	0.8091	0.8868	0.9928	0.5940	0.8592	0.9584	0.2700	0.6233	0.7248
	3(20)	0.7377	0.5204	0.9826	0.5562	0.9409	0.9408	0.2027	0.2160	0.5991
	3(30)	0.8223	0.8099	0.9945	0.7303	0.8737	0.9847	0.3865	0.2319	0.8555
	3(40)	0.8708	0.6780	0.9972	0.7734	0.4909	0.9910	0.3779	0.3970	0.8459
	3(50)	0.9224	0.7355	0.9990	0.7905	0.6223	0.9919	0.3531	0.2217	0.8134
	3(5)	0.2211	0.3024	0.7124	0.1556	0.3784	0.5683	0.1178	0.1877	0.4366
	3(10)	0.8584	0.8656	0.9964	0.7903	0.8656	0.9929	0.5054	0.5037	0.9421
	3(15)	0.9010	0.8868	0.9983	0.6999	0.8868	0.9809	0.2751	0.6510	0.7330
$\Delta=0.75$	3(20)	0.9017	0.7755	0.9971	0.7489	0.5916	0.9846	0.2655	0.2351	0.7091
	3(30)	0.9777	0.9616	0.9997	0.9371	0.8810	0.9988	0.5885	0.3970	0.9599
	3(40)	0.9899	0.9589	0.9999	0.9615	0.8656	0.9995	0.5944	0.4909	0.9594
	3(50)	0.9957	0.8907	1.0000	0.9634	0.8002	0.9998	0.5460	0.4900	0.9432
	3(5)	0.2966	0.3784	0.8230	0.1810	0.3024	0.6590	0.0928	0.1507	0.3193
	3(10)	0.9538	0.8656	0.9994	0.9081	0.8656	0.9988	0.6261	0.5991	0.9785
	3(15)	0.9668	0.8937	0.9998	0.8253	0.8868	0.9941	0.3084	0.6938	0.7741
	3(20)	0.9808	0.9599	0.9998	0.9053	0.6938	0.9974	0.3684	0.2351	0.8349
	3(30)	0.9992	0.9870	1.0000	0.9936	0.9689	1.0000	0.7739	0.5529	0.9906
	3(40)	0.9998	0.9944	1.0000	0.9979	0.9215	1.0000	0.7976	0.5991	0.9924
$\Delta=1.50$	3(50)	1.0000	0.9888	1.0000	0.9980	0.9071	1.0000	0.7500	0.6763	0.9875
	3(5)	0.5231	0.5201	0.9430	0.3371	0.3784	0.8567	0.0816	0.1507	0.2537
	3(10)	0.9983	0.9165	1.0000	0.9918	0.8656	1.0000	0.8279	0.8312	0.9975
	3(15)	0.9992	0.9623	1.0000	0.9762	0.9065	0.9997	0.4548	0.7399	0.9031
	3(20)	0.9999	0.9891	1.0000	0.9971	0.9764	1.0000	0.6491	0.3311	0.9696
	3(30)	1.0000	0.9998	1.0000	1.0000	0.9983	1.0000	0.9666	0.8312	0.9996
	3(40)	1.0000	1.0000	1.0000	1.0000	0.9985	1.0000	0.9813	0.8134	0.9999
	3(50)	1.0000	1.0000	1.0000	1.0000	0.9753	1.0000	0.9703	0.8579	1.0000

Table 4 Power of F-Test, FR-Test and R-Test under normal error for $t=6$

Effect sizes	$t(b)$	$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1$			$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1.5$			$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 2 : 3$		
		F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test
$\Delta=0.50$	6(5)	0.5273	0.6364	0.8767	0.5176	0.6364	0.8702	0.4147	0.5904	0.7852
	6(10)	0.5529	0.6284	0.8753	0.4973	0.6316	0.8220	0.2592	0.2243	0.4577
	6(15)	0.8373	0.8625	0.9884	0.7580	0.8625	0.9720	0.4229	0.7427	0.7416
	6(20)	0.7913	0.7720	0.9767	0.7417	0.7386	0.9613	0.3598	0.3262	0.6340
	6(30)	0.5059	0.6824	0.8180	0.4014	0.6525	0.6962	0.1489	0.2165	0.1642
	6(40)	0.4531	0.4854	0.7623	0.3295	0.3899	0.5710	0.1295	0.1825	0.1167
$\Delta=0.75$	6(50)	0.7163	0.6739	0.9502	0.5856	0.5445	0.8801	0.1803	0.2134	0.2366
	6(5)	0.6064	0.6904	0.9195	0.5974	0.6671	0.9146	0.4524	0.6427	0.8186
	6(10)	0.7192	0.7112	0.9568	0.6603	0.6535	0.9342	0.3317	0.2877	0.5991
	6(15)	0.8952	0.8941	0.9949	0.8156	0.8645	0.9838	0.4171	0.7223	0.7358
	6(20)	0.9345	0.8642	0.9975	0.9050	0.7709	0.9943	0.4920	0.4564	0.8056
	6(30)	0.7737	0.7784	0.9736	0.6444	0.7577	0.9215	0.1923	0.2191	0.2807
$\Delta=1.00$	6(40)	0.7496	0.7258	0.9616	0.5513	0.4921	0.8512	0.1320	0.1429	0.1192
	6(50)	0.9466	0.8859	0.9983	0.8665	0.7649	0.9901	0.2775	0.3290	0.4530
	6(5)	0.7016	0.6904	0.9560	0.6930	0.6904	0.9521	0.5012	0.6671	0.8589
	6(10)	0.8661	0.8243	0.9899	0.8156	0.7030	0.9817	0.4221	0.2877	0.7335
	6(15)	0.9561	0.9682	0.9983	0.8989	0.8941	0.9945	0.4423	0.7258	0.7686
	6(20)	0.9904	0.9480	0.9999	0.9810	0.8845	0.9996	0.6419	0.5430	0.9149
$\Delta=1.50$	6(30)	0.9582	0.9677	0.9996	0.8912	0.8149	0.9940	0.2929	0.2837	0.5093
	6(40)	0.9633	0.9505	0.998	0.8555	0.7051	0.9864	0.1957	0.1508	0.2768
	6(50)	0.9983	0.9761	1.0000	0.9884	0.9223	0.9999	0.4632	0.4926	0.7566
	6(5)	0.8793	0.7232	0.9888	0.8730	0.7232	0.9876	0.6234	0.6904	0.9244
	6(10)	0.9894	0.9336	0.9996	0.9766	0.8715	0.9992	0.6333	0.4329	0.9192
	6(15)	0.9985	0.9947	1.0000	0.9917	0.9713	1.0000	0.5767	0.7292	0.8907
$\Delta=1.50$	6(20)	1.0000	0.9974	1.0000	1.0000	0.9782	1.0000	0.8877	0.7411	0.9919
	6(30)	1.0000	0.9992	1.0000	0.9994	0.9870	1.0000	0.6405	0.4421	0.9154
	6(40)	1.0000	0.9999	1.0000	0.9996	0.9877	1.0000	0.5360	0.4200	0.8270
	6(50)	1.0000	0.9999	1.0000	1.0000	0.9996	1.0000	0.8758	0.7695	0.9903

Table 5 Power of F-Test, FR-Test and R-Test under normal error for $t=9$

Effect sizes	$t(b)$	$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1$			$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1.5$			$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 2 : 3$		
		F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test
$\Delta=0.50$	9(5)	0.4272	0.5195	0.6325	0.3918	0.4610	0.5735	0.3431	0.4483	0.4868
	9(10)	0.7303	0.8631	0.9192	0.6668	0.8248	0.8754	0.4922	0.7672	0.6809
	9(15)	0.8253	0.8143	0.9633	0.7414	0.8122	0.9190	0.3836	0.4910	0.4683
	9(20)	0.5750	0.6009	0.7529	0.4857	0.5119	0.6264	0.2173	0.2422	0.1281
	9(30)	0.7155	0.8894	0.8960	0.7750	0.8543	0.9368	0.4545	0.3396	0.5603
	9(40)	0.5234	0.5531	0.6770	0.4350	0.4809	0.5288	0.2305	0.2239	0.1479
	9(50)	0.6331	0.6572	0.8070	0.5453	0.6744	0.6983	0.1658	0.2481	0.0568
	$\Delta=0.75$	0.4333	0.5579	0.6393	0.3704	0.5104	0.5359	0.3109	0.4101	0.4194
		0.8069	0.8758	0.9588	0.7476	0.8397	0.9289	0.5210	0.7643	0.7196
		0.9053	0.8936	0.9883	0.8417	0.8621	0.9689	0.4349	0.5275	0.5606
		0.7131	0.7022	0.8991	0.5987	0.5679	0.7840	0.2235	0.2179	0.1470
		0.9401	0.9758	0.9953	0.9597	0.9695	0.9980	0.6482	0.6043	0.8354
$\Delta=1.00$	9(40)	0.7821	0.8102	0.9361	0.6527	0.6036	0.8315	0.2439	0.2538	0.1739
	9(50)	0.9114	0.9188	0.9890	0.8439	0.7848	0.9674	0.2474	0.3694	0.1767
	9(5)	0.4710	0.5835	0.6936	0.3771	0.5074	0.5475	0.2880	0.4357	0.3699
	9(10)	0.8915	0.9138	0.9861	0.8438	0.8705	0.9727	0.5654	0.7691	0.7721
	9(15)	0.9671	0.9450	0.9989	0.9316	0.8986	0.9944	0.5093	0.5675	0.6802
	9(20)	0.8776	0.8470	0.9817	0.7770	0.6617	0.9404	0.2616	0.2171	0.2135
	9(30)	0.9962	0.9974	1.0000	0.9977	0.9930	1.0000	0.8204	0.7434	0.9618
	9(40)	0.9674	0.9626	0.9982	0.9081	0.8539	0.9882	0.3252	0.2845	0.3210
	9(50)	0.9962	0.9943	1.0000	0.9871	0.9435	0.9999	0.4219	0.4518	0.4972
	$\Delta=1.50$	0.6275	0.6191	0.8577	0.4740	0.5722	0.6996	0.2697	0.4672	0.3292
		0.9867	0.9744	0.9996	0.9731	0.9297	0.9984	0.6892	0.7928	0.8894
		0.9993	0.9789	1.0000	0.9968	0.9692	1.0000	0.7015	0.6367	0.8881
		0.9971	0.9731	1.0000	0.9858	0.9211	0.9999	0.4404	0.3495	0.5483
		1.0000	1.0000	1.0000	1.0000	0.9997	1.0000	0.9806	0.9338	0.9993
$\Delta=1.50$	9(40)	1.0000	0.9998	1.0000	0.9999	0.9949	1.0000	0.6674	0.5586	0.8451
	9(50)	1.0000	1.0000	1.0000	1.0000	0.9997	1.0000	0.8602	0.7479	0.9719

Table 6 Type-I error rate of F-Test, FR-Test and R-Test under skewed error (Cauchy)

Number of treatments		$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1$			$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1.5$			$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 2 : 3$		
b		F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test
$t=3$	5	0.0167	0.0189	0.0189	0.0177	0.0285	0.0285	0.0200	0.0187	0.0187
	10	0.0167	0.0177	0.0177	0.0179	0.0023	0.0023	0.0210	0.0000	0.0000
	15	0.0167	0.0000	0.0000	0.0164	0.0000	0.0000	0.0196	0.0000	0.0000
	20	0.0171	0.0148	0.0148	0.0188	0.0035	0.0035	0.0209	0.0002	0.0002
	30	0.0151	0.0000	0.0000	0.0160	0.0000	0.0000	0.0193	0.0000	0.0000
	40	0.0156	0.0458	0.0458	0.0163	0.0458	0.0458	0.0195	0.0443	0.0443
$t=6$	50	0.0139	0.0344	0.0344	0.0162	0.0284	0.0284	0.0191	0.0216	0.0216
	5	0.0188	0.0047	0.0047	0.0162	0.0341	0.0341	0.0165	0.0320	0.0320
	10	0.0156	0.0221	0.0221	0.0158	0.0065	0.0065	0.0178	0.0070	0.0070
	15	0.0170	0.0017	0.0017	0.0163	0.0194	0.0194	0.0188	0.0199	0.0199
	20	0.0181	0.0027	0.0027	0.0160	0.0027	0.0027	0.0178	0.0024	0.0024
	30	0.0184	0.0133	0.0133	0.0171	0.0051	0.0051	0.0193	0.0050	0.0050
$t=9$	40	0.0154	0.0346	0.0346	0.0166	0.0155	0.0155	0.0182	0.0156	0.0156
	50	0.0146	0.0251	0.0251	0.0169	0.0052	0.0052	0.0182	0.0117	0.0117
	5	0.0188	0.0047	0.0047	0.0180	0.0053	0.0053	0.0199	0.0034	0.0034
	10	0.0156	0.0221	0.0221	0.0162	0.0225	0.0225	0.0170	0.0216	0.0216
	15	0.0170	0.0017	0.0017	0.0176	0.0021	0.0021	0.0186	0.0021	0.0021
	20	0.0181	0.0027	0.0027	0.0192	0.0026	0.0026	0.0209	0.0028	0.0028
	30	0.0184	0.0133	0.0133	0.0200	0.0148	0.0148	0.0210	0.0150	0.0150
	40	0.0154	0.0346	0.0346	0.0168	0.0346	0.0346	0.0184	0.0339	0.0339
	50	0.0146	0.0251	0.0251	0.0156	0.0259	0.0259	0.0170	0.0303	0.0303

Table 7 Power of F-Test, FR-Test and R-Test under skewed error (Cauchy) for $t=3$

Effect sizes	$t(b)$	$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1$			$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1.5$			$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 2 : 3$		
		F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test
$\Delta=0.50$	3(5)	0.7207	0.4866	0.9793	0.7876	0.4867	0.9857	0.7611	0.3024	0.9836
	3(10)	0.1840	0.5037	0.4320	0.1792	0.4362	0.3364	0.1845	0.2831	0.3895
	3(15)	0.1545	0.0814	0.2191	0.1446	0.0814	0.2201	0.1374	0.1268	0.1776
	3(20)	0.1669	0.1065	0.3132	0.1546	0.1065	0.3062	0.1453	0.0814	0.2453
	3(30)	0.1387	0.1209	0.1947	0.1298	0.2319	0.1873	0.0940	0.1691	0.1700
	3(40)	0.4746	0.2113	0.9680	0.4565	0.2113	0.9706	0.4343	0.0122	0.9693
	3(50)	0.2775	0.2754	0.7246	0.2623	0.2754	0.7055	0.2234	0.1728	0.5963
	3(5)	0.8180	0.4866	0.9891	0.8525	0.4866	0.9900	0.7985	0.3024	0.9865
	3(10)	0.1733	0.4336	0.3450	0.1719	0.5037	0.2668	0.1808	0.2831	0.3490
	3(15)	0.1596	0.0814	0.2487	0.1478	0.0814	0.2343	0.1387	0.0814	0.1822
$\Delta=0.75$	3(20)	0.1724	0.0814	0.3366	0.1566	0.0733	0.3133	0.1460	0.0814	0.2474
	3(30)	0.1618	0.2831	0.3023	0.1507	0.2831	0.2832	0.1098	0.3088	0.2383
	3(40)	0.5014	0.2113	0.9838	0.4749	0.2113	0.9816	0.4437	0.2208	0.9760
	3(50)	0.3018	0.2217	0.7639	0.2802	0.2319	0.7516	0.2318	0.1728	0.6282
	3(5)	0.8908	0.4866	0.9945	0.9025	0.4866	0.9935	0.8322	0.3024	0.9887
	3(10)	0.1630	0.4336	0.2675	0.1650	0.4336	0.2042	0.1772	0.2831	0.3118
	3(15)	0.1656	0.0601	0.2864	0.1519	0.0814	0.2575	0.1403	0.0601	0.1917
	3(20)	0.1797	0.2351	0.3780	0.1603	0.0814	0.3297	0.1471	0.0814	0.2512
	3(30)	0.1893	0.5259	0.4392	0.1759	0.3659	0.3984	0.1286	0.3659	0.3153
	3(40)	0.5282	0.5529	0.9920	0.4936	0.2687	0.9890	0.4531	0.2831	0.9818
$\Delta=1.00$	3(50)	0.3276	0.2677	0.7975	0.2989	0.2217	0.7953	0.2405	0.1116	0.6595
	3(5)	0.9696	0.4866	0.9974	0.9635	0.4866	0.9964	0.8879	0.4866	0.9915
	3(10)	0.1441	0.4336	0.1436	0.1518	0.4336	0.1090	0.1700	0.2065	0.2457
	3(15)	0.1809	0.1278	0.3802	0.1632	0.0924	0.3215	0.1443	0.0601	0.2113
	3(20)	0.1999	0.3784	0.4956	0.1727	0.3784	0.3895	0.1505	0.0500	0.2662
	3(30)	0.2577	0.8071	0.7163	0.2394	0.8071	0.6490	0.1759	0.4574	0.4754
	3(40)	0.5817	0.6907	0.9989	0.5309	0.4108	0.9984	0.4720	0.3358	0.9903
	3(50)	0.3828	0.3821	0.8713	0.3385	0.2217	0.8703	0.2585	0.1186	0.7196

Table 8 Power of F-Test, FR-Test and R-Test under skewed error (Cauchy) for $t=6$

Effect sizes	$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1$			$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1.5$			$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 2 : 3$		
	F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test
$\Delta=0.50$	$t(6)$								
	6(5)	0.2343	0.4349	0.2415	0.2340	0.4349	0.1844	0.2377	0.3211
	6(10)	0.2566	0.3607	0.3121	0.2467	0.3201	0.2993	0.2484	0.2717
	6(15)	0.2193	0.1298	0.1575	0.2134	0.1187	0.1469	0.2071	0.1389
	6(20)	0.4223	0.1412	0.7672	0.4239	0.1918	0.7906	0.4199	0.1974
	6(30)	0.5167	0.5633	0.8719	0.4345	0.5418	0.7316	0.4475	0.5514
$\Delta=0.75$	6(40)	0.5707	0.1025	0.9316	0.5635	0.1033	0.9217	0.5762	0.1334
	6(50)	0.4112	0.3177	0.6769	0.4063	0.2596	0.6645	0.4085	0.2605
	6(5)	0.2258	0.4349	0.2058	0.2284	0.4349	0.1589	0.2338	0.3221
	6(10)	0.2591	0.3809	0.3245	0.2481	0.3445	0.3026	0.2488	0.2359
	6(15)	0.2253	0.1624	0.1725	0.2178	0.1187	0.1604	0.2088	0.1435
	6(20)	0.4391	0.1588	0.7963	0.4367	0.1678	0.8111	0.4263	0.2050
$\Delta=1.00$	6(30)	0.5083	0.6784	0.8548	0.4233	0.6111	0.7034	0.4415	0.6306
	6(40)	0.5717	0.1624	0.9332	0.5641	0.1326	0.9219	0.5759	0.1335
	6(50)	0.4125	0.5321	0.6781	0.4056	0.4850	0.6632	0.4072	0.2653
	6(5)	0.2176	0.4349	0.1741	0.2228	0.4349	0.1396	0.2309	0.3221
	6(10)	0.2625	0.3809	0.3396	0.2501	0.3809	0.3094	0.2493	0.2359
	6(15)	0.2331	0.1918	0.1977	0.2239	0.1576	0.1789	0.2109	0.1435
$\Delta=1.50$	6(20)	0.4561	0.1955	0.8215	0.4496	0.1732	0.8271	0.4327	0.2012
	6(30)	0.4934	0.7593	0.8390	0.4124	0.6616	0.6748	0.4356	0.6463
	6(40)	0.5734	0.1974	0.9348	0.5655	0.2330	0.9235	0.5759	0.1421
	6(50)	0.4143	0.7626	0.6819	0.4062	0.6952	0.6632	0.4062	0.3525
	6(5)	0.2019	0.4349	0.1226	0.2119	0.3708	0.1084	0.2253	0.3221
	6(10)	0.2715	0.3647	0.3813	0.2565	0.3809	0.3345	0.2509	0.3114
	6(15)	0.2541	0.5847	0.2725	0.2415	0.4150	0.2384	0.2163	0.1518
	6(20)	0.4909	0.3728	0.8592	0.4757	0.2031	0.8613	0.4456	0.2088
	6(30)	0.4715	0.9190	0.8019	0.3913	0.8249	0.6108	0.4239	0.6889
	6(40)	0.5792	0.5202	0.9389	0.5704	0.4378	0.9273	0.5765	0.2022
	6(50)	0.4222	0.9816	0.6982	0.4115	0.8967	0.6760	0.4053	0.5763
									0.6568

Table 9 Power of F-Test, FR-Test and R-Test under skewed error (Cauchy) for $t=9$

Effect sizes	$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1$				$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1.5$				$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 2 : 3$			
	$t(b)$	F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test	F-Test	R-Test
$\Delta = 0.50$	9(5)	0.3210	0.4101	0.2766	0.3207	0.4101	0.3025	0.3212	0.3552	0.3040	0.3212	0.3040
	9(10)	0.3489	0.4133	0.2686	0.3447	0.4277	0.2677	0.3473	0.3552	0.2742	0.3473	0.2742
	9(15)	0.4759	0.3897	0.6641	0.4825	0.5354	0.6880	0.4833	0.5165	0.6909	0.4833	0.6909
	9(20)	0.4908	0.4944	0.5715	0.4541	0.4602	0.4367	0.4542	0.4882	0.4357	0.4542	0.4357
	9(30)	0.6552	0.5896	0.9074	0.6561	0.5472	0.9082	0.6673	0.6174	0.9351	0.6673	0.9351
	9(40)	0.6995	0.6681	0.9484	0.7011	0.6346	0.9500	0.4437	0.2207	0.9760	0.4437	0.9760
	9(50)	0.5089	0.3056	0.6079	0.5148	0.2998	0.6233	0.7121	0.5981	0.9684	0.7121	0.9684
	9(5)	0.3199	0.3875	0.2673	0.3183	0.3875	0.2895	0.3212	0.3552	0.3040	0.3212	0.3040
	9(10)	0.3528	0.4293	0.2820	0.3475	0.4531	0.2775	0.3473	0.3552	0.2742	0.3473	0.2742
	9(15)	0.4731	0.3315	0.6536	0.4789	0.4314	0.6740	0.4833	0.5164	0.6909	0.4833	0.6909
$\Delta = 0.75$	9(20)	0.4798	0.5608	0.5391	0.4439	0.4943	0.4024	0.4542	0.4882	0.4357	0.4542	0.4357
	9(30)	0.6574	0.6879	0.9096	0.6587	0.6960	0.9100	0.6673	0.6174	0.9351	0.6673	0.9351
	9(40)	0.7031	0.7835	0.9510	0.7054	0.7335	0.9525	0.7138	0.6841	0.9695	0.7138	0.9695
	9(50)	0.5032	0.3425	0.5936	0.5053	0.3294	0.6027	0.5154	0.2826	0.6209	0.5154	0.6209
	9(5)	0.3190	0.3617	0.2613	0.3161	0.3875	0.2772	0.3199	0.3552	0.2969	0.3199	0.2969
	9(10)	0.3577	0.5209	0.2998	0.3515	0.4531	0.2899	0.3486	0.3827	0.2774	0.3486	0.2774
	9(15)	0.4706	0.2500	0.6437	0.4758	0.4133	0.6607	0.4815	0.4704	0.6847	0.4815	0.6847
	9(20)	0.4690	0.5968	0.5067	0.4339	0.5395	0.3774	0.4491	0.5217	0.4172	0.4491	0.4172
	9(30)	0.6601	0.7899	0.9125	0.6619	0.7515	0.9129	0.6685	0.6847	0.9363	0.6685	0.9363
	9(40)	0.7075	0.8963	0.9543	0.7104	0.8757	0.9555	0.7157	0.7252	0.9701	0.7157	0.9701
$\Delta = 1.50$	9(50)	0.4991	0.4591	0.5821	0.4973	0.3875	0.5855	0.5099	0.2712	0.6089	0.5099	0.6089
	9(5)	0.3180	0.3552	0.2539	0.3125	0.3552	0.2573	0.3176	0.3358	0.2854	0.3176	0.2854
	9(10)	0.3709	0.5447	0.3457	0.3628	0.5164	0.3283	0.3519	0.3827	0.2899	0.3519	0.2899
	9(15)	0.4665	0.1966	0.6234	0.4704	0.2114	0.6399	0.4782	0.4026	0.6726	0.4782	0.6726
	9(20)	0.4481	0.7459	0.4370	0.4144	0.6554	0.3290	0.4391	0.5359	0.3865	0.4391	0.3865
	9(30)	0.6670	0.9727	0.9203	0.6698	0.9317	0.9223	0.6712	0.8122	0.9385	0.6712	0.9385
	9(40)	0.7186	0.9887	0.9612	0.7227	0.9687	0.9637	0.7201	0.8534	0.9721	0.7201	0.9721
	9(50)	0.4954	0.8019	0.5730	0.4858	0.5841	0.5602	0.5002	0.3290	0.5851	0.5002	0.5851

Table 10 Type-I-error rate of F-Test, FR-Test and R-Test under skewed error (logistic)

Number of treatments		$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1$			$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1.5$			$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 2 : 3$		
t	b	F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test
$t=3$	5	0.0034	0.0519	0.0519	0.0042	0.0452	0.0452	0.0046	0.0419	0.0419
	10	0.0014	0.0465	0.0465	0.0020	0.0458	0.0458	0.0027	0.0441	0.0441
	15	0.0010	0.0487	0.0487	0.0019	0.0505	0.0505	0.0029	0.0480	0.0480
	20	0.0010	0.0504	0.0504	0.0018	0.0504	0.0504	0.0030	0.0503	0.0503
	30	0.0009	0.0493	0.0493	0.0013	0.0491	0.0491	0.0024	0.0501	0.0501
	40	0.0013	0.0492	0.0492	0.0021	0.0498	0.0498	0.0036	0.0511	0.0511
	50	0.0009	0.0507	0.0507	0.0015	0.0494	0.0494	0.0031	0.0500	0.0500
	5	0.0503	0.0424	0.0424	0.0571	0.0435	0.0435	0.0629	0.0428	0.0428
	10	0.0492	0.0407	0.0407	0.0542	0.0436	0.0436	0.0591	0.0432	0.0432
	15	0.0522	0.0450	0.0450	0.0560	0.0494	0.0494	0.0644	0.0468	0.0468
$t=6$	20	0.0504	0.0402	0.0402	0.0580	0.0430	0.0430	0.0626	0.0447	0.0447
	30	0.0498	0.0414	0.0414	0.0583	0.0430	0.0430	0.0640	0.0430	0.0430
	40	0.0527	0.0379	0.0379	0.0563	0.0382	0.0382	0.0634	0.0401	0.0401
	50	0.0510	0.0467	0.0467	0.0561	0.0501	0.0501	0.0629	0.0492	0.0492
	5	0.0528	0.0411	0.0411	0.0569	0.0355	0.0355	0.0635	0.0357	0.0357
	10	0.0496	0.0442	0.0442	0.0527	0.0445	0.0445	0.0579	0.0436	0.0436
	15	0.0484	0.0467	0.0467	0.0548	0.0489	0.0489	0.0650	0.0481	0.0481
	20	0.0515	0.0433	0.0433	0.0584	0.0453	0.0453	0.0628	0.0460	0.0460
	30	0.0504	0.0455	0.0455	0.0553	0.0450	0.0450	0.0613	0.0462	0.0462
	40	0.0527	0.0471	0.0471	0.0566	0.0473	0.0473	0.0635	0.0492	0.0492
$t=9$	50	0.0494	0.0437	0.0437	0.0544	0.0439	0.0439	0.0622	0.0452	0.0452

Table 11 Power of F-Test, FR-Test and R-Test under skewed error (logistic) for $t = 3$

Effect sizes	$r(b)$	$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1$				$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1.5$				$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 2 : 3$			
		F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test
$\Delta = 0.50$	3(5)	0.4005	0.5204	0.8851	0.3459	0.5204	0.8492	0.4725	0.4866	0.9254	0.4725	0.4866	0.9254
	3(10)	0.4042	0.5529	0.8777	0.2971	0.5529	0.7789	0.1070	0.2065	0.3348	0.1070	0.2065	0.3348
	3(15)	0.9495	0.6233	0.9990	0.9629	0.8208	0.9992	0.9484	0.7252	0.9992	0.9484	0.7252	0.9992
	3(20)	0.7409	0.9058	0.9850	0.5989	0.7849	0.9580	0.2847	0.4952	0.7414	0.2847	0.4952	0.7414
	3(30)	0.8871	0.8656	0.9976	0.7643	0.6729	0.9900	0.4173	0.4396	0.8764	0.4173	0.4396	0.8764
	3(40)	0.5491	0.3263	0.9364	0.4100	0.2208	0.8626	0.2042	0.0980	0.5967	0.2042	0.0980	0.5967
	3(50)	0.9521	0.8705	0.9994	0.9622	0.9229	0.9995	0.8990	0.8105	0.9982	0.8990	0.8105	0.9982
	3(5)	0.2211	0.3024	0.7124	0.1556	0.3784	0.5683	0.4948	0.4866	0.9343	0.4948	0.4866	0.9343
	3(10)	0.8584	0.8656	0.9964	0.7903	0.8656	0.9929	0.1097	0.2065	0.3442	0.1097	0.2065	0.3442
	3(15)	0.9010	0.8868	0.9983	0.6999	0.8868	0.9809	0.9688	0.7252	0.9994	0.9688	0.7252	0.9994
$\Delta = 0.75$	3(20)	0.9017	0.7755	0.9971	0.7489	0.5916	0.9846	0.3343	0.4426	0.8002	0.3343	0.4426	0.8002
	3(30)	0.9777	0.9616	0.9997	0.9371	0.8850	0.9988	0.4634	0.4396	0.9052	0.4634	0.4396	0.9052
	3(40)	0.9899	0.9589	0.9999	0.9615	0.8656	0.9995	0.2286	0.1283	0.6452	0.2286	0.1283	0.6452
	3(50)	0.9899	0.9589	0.9999	0.9957	0.8907	1.0000	0.9544	0.8429	0.9993	0.9544	0.8429	0.9993
	3(5)	0.5034	0.5204	0.9302	0.4369	0.5204	0.9026	0.5187	0.4866	0.9421	0.5187	0.4866	0.9421
	3(10)	0.5723	0.5368	0.9537	0.4157	0.5529	0.8855	0.1208	0.2065	0.3907	0.1208	0.2065	0.3907
	3(15)	0.9938	0.8937	0.9998	0.9944	0.8207	0.9999	0.9820	0.8208	0.9995	0.9820	0.8208	0.9995
	3(20)	0.9082	0.9058	0.9982	0.7984	0.8492	0.9922	0.3946	0.5204	0.8555	0.3946	0.5204	0.8555
	3(30)	0.9753	0.9256	0.9999	0.9147	0.8656	0.9985	0.5239	0.4923	0.9358	0.5239	0.4923	0.9358
	3(40)	0.8656	0.4736	0.9955	0.7187	0.6521	0.9806	0.2813	0.1553	0.7338	0.2813	0.1553	0.7338
$\Delta = 1.50$	3(50)	0.9995	0.9505	1.0000	0.9993	0.9473	1.0000	0.9824	0.8429	0.9999	0.9824	0.8429	0.9999
	3(5)	0.6184	0.7399	0.9584	0.5446	0.5200	0.9419	0.5702	0.5204	0.9556	0.5702	0.5204	0.9556
	3(10)	0.7611	0.6686	0.9864	0.5905	0.5529	0.9587	0.1700	0.2831	0.5472	0.1700	0.2831	0.5472
	3(15)	0.9996	0.9841	1.0000	0.9995	0.8937	1.0000	0.9948	0.8937	0.9998	0.9948	0.8937	0.9998
	3(20)	0.9843	0.9599	0.9998	0.9369	0.9599	0.9993	0.5386	0.6877	0.9399	0.5386	0.6877	0.9399
	3(30)	0.9984	0.9468	1.0000	0.9863	0.9359	1.0000	0.6716	0.6729	0.9757	0.6716	0.6729	0.9757
	3(40)	0.9921	0.9605	1.0000	0.9539	0.8071	0.9997	0.4649	0.2065	0.9015	0.4649	0.2065	0.9015
	3(50)	1.0000	0.9954	1.0000	1.0000	0.9869	1.0000	0.9984	0.8843	1.0000	0.9984	0.8843	1.0000

Table 12 Power of F-Test, FR-Test and R-Test under skewed error (logistic) for $t=6$

Effect sizes	$t(b)$	$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1$			$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1.5$			$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 2 : 3$		
		F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test	F-Test	FR-Test	R-Test
$\Delta=0.50$	6(5)	0.4444	0.4110	0.8140	0.4075	0.4110	0.7735	0.5392	0.3789	0.8909
	6(10)	0.8356	0.8206	0.9885	0.8285	0.8168	0.9877	0.6563	0.7398	0.9398
	6(15)	0.8284	0.6247	0.9880	0.7242	0.5520	0.9629	0.5262	0.5915	0.8487
	6(20)	0.8099	0.8280	0.9829	0.7691	0.8224	0.9727	0.4814	0.5644	0.7977
	6(30)	0.9240	0.8442	0.9980	0.8710	0.8286	0.9928	0.6287	0.7976	0.9143
	6(40)	0.4618	0.5257	0.7726	0.3747	0.5294	0.6489	0.2419	0.4230	0.3909
	6(50)	0.7160	0.5880	0.9545	0.5892	0.6002	0.8874	0.4147	0.5666	0.7041
	6(5)	0.4761	0.4110	0.8439	0.4307	0.4110	0.7986	0.5476	0.3789	0.8947
$\Delta=0.75$	6(10)	0.8779	0.8369	0.9943	0.8751	0.8168	0.9933	0.6976	0.7497	0.9539
	6(15)	0.8628	0.7477	0.9923	0.7662	0.5662	0.9749	0.5429	0.5544	0.8641
	6(20)	0.8817	0.8627	0.9931	0.8522	0.8437	0.9891	0.5462	0.5644	0.8558
	6(30)	0.9356	0.8519	0.9986	0.8785	0.8298	0.9936	0.5990	0.7716	0.8977
	6(40)	0.5388	0.6512	0.8485	0.4086	0.5618	0.6981	0.2149	0.4010	0.3305
	6(50)	0.8335	0.7312	0.9854	0.7155	0.5975	0.9522	0.4547	0.5517	0.7573
	6(5)	0.5167	0.3950	0.8712	0.4632	0.4110	0.8288	0.5588	0.3789	0.9008
	6(10)	0.9168	0.8627	0.9967	0.9160	0.8386	0.9973	0.7397	0.7497	0.9685
$\Delta=1.00$	6(15)	0.9025	0.7525	0.9960	0.8181	0.5961	0.9861	0.5676	0.5847	0.8821
	6(20)	0.9396	0.8771	0.9983	0.9200	0.8589	0.9964	0.6175	0.5921	0.9057
	6(30)	0.9588	0.8885	0.9993	0.9078	0.8369	0.9960	0.5866	0.7739	0.8898
	6(40)	0.6912	0.6974	0.9436	0.5241	0.6228	0.8316	0.2115	0.4070	0.3213
	6(50)	0.9366	0.8202	0.9978	0.8543	0.7101	0.9883	0.5193	0.5409	0.8226
	6(5)	0.6187	0.5556	0.9253	0.5527	0.4506	0.8925	0.5891	0.3869	0.9165
	6(10)	0.9710	0.8864	0.9998	0.9710	0.8926	0.9998	0.8209	0.7969	0.9863
	6(15)	0.9673	0.8120	0.9995	0.9185	0.7007	0.9969	0.6366	0.6289	0.9243
$\Delta=1.50$	6(20)	0.9923	0.9558	1.0000	0.9871	0.9260	1.0000	0.7616	0.6656	0.9690
	6(30)	0.9936	0.9818	0.9999	0.9750	0.9405	0.9995	0.6160	0.8200	0.9061
	6(40)	0.9542	0.9199	0.9982	0.8520	0.8051	0.9874	0.2761	0.4269	0.4636
	6(50)	0.9983	0.9924	1.0000	0.9888	0.8365	0.9996	0.6981	0.6322	0.9446

Table 13 Power of F-Test, FR-Test and R-Test under skewed error (logistic) for $t=9$

Effect sizes		$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1$				$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 1 : 1.5$				$\sigma_1^2 : \sigma_2^2 : \sigma_3^2 = 1 : 2 : 3$			
		F-Test	FR-Test	R-Test		F-Test	FR-Test	R-Test		F-Test	FR-Test	R-Test	
$\Delta=0.50$	$t(5)$	0.8123	0.7485	0.9646		0.7834	0.7255	0.9563		0.8208	0.7866	0.9695	
	$t(10)$	0.7589	0.7382	0.9395		0.6955	0.7033	0.8978		0.5239	0.7090	0.7195	
	$t(15)$	0.8835	0.8028	0.9836		0.8888	0.8667	0.9836		0.7291	0.8122	0.9137	
	$t(20)$	0.9010	0.9116	0.9871		0.8690	0.9167	0.9779		0.6654	0.8528	0.8598	
	$t(30)$	0.8843	0.8721	0.9828		0.7995	0.8382	0.9471		0.5363	0.7131	0.6995	
	$t(40)$	0.3917	0.4297	0.4516		0.3620	0.4137	0.3959		0.2694	0.3532	0.2212	
	$t(50)$	0.6407	0.7543	0.8258		0.5580	0.7989	0.7245		0.3828	0.7148	0.4382	
	$t(5)$	0.8426	0.7485	0.9750		0.8194	0.7255	0.9683		0.8370	0.7866	0.9754	
	$t(10)$	0.7576	0.7485	0.9391		0.6814	0.7329	0.8855		0.5005	0.7010	0.5005	
	$t(15)$	0.9280	0.8348	0.9926		0.9349	0.8687	0.9945		0.7788	0.8373	0.9426	
$\Delta=0.75$	$t(20)$	0.9271	0.9386	0.9927		0.9025	0.9240	0.9876		0.6870	0.8750	0.8774	
	$t(30)$	0.9230	0.8752	0.9912		0.8547	0.8615	0.9721		0.5661	0.7308	0.7428	
	$t(40)$	0.4327	0.4535	0.5246		0.3543	0.4133	0.3812		0.2283	0.3072	0.1481	
	$t(50)$	0.7755	0.9134	0.9310		0.6945	0.8709	0.8714		0.4279	0.7745	0.5215	
	$t(5)$	0.8724	0.7549	0.9829		0.8539	0.7298	0.9786		0.8530	0.7866	0.9798	
	$t(10)$	0.7694	0.7575	0.9451		0.6811	0.7555	0.8862		0.4821	0.6930	0.6597	
	$t(15)$	0.9617	0.8617	0.9979		0.9669	0.8936	0.9981		0.8254	0.8392	0.9627	
	$t(20)$	0.9555	0.9476	0.9971		0.9386	0.9196	0.9940		0.7170	0.8726	0.9020	
	$t(30)$	0.9602	0.9204	0.9978		0.9130	0.8819	0.9887		0.6077	0.7498	0.7920	
	$t(40)$	0.5583	0.4948	0.7216		0.4277	0.4428	0.5135		0.2087	0.2992	0.1149	
$\Delta=1.00$	$t(50)$	0.9053	0.9699	0.9882		0.8452	0.9236	0.9664		0.4970	0.8172	0.6341	
	$t(5)$	0.9253	0.8548	0.9924		0.9135	0.8039	0.9916		0.8835	0.8073	0.9875	
	$t(10)$	0.8252	0.8474	0.967		0.7208	0.7784	0.9139		0.4615	0.7112	0.6302	
	$t(15)$	0.9936	0.9545	0.9998		0.9947	0.9282	0.9999		0.9028	0.8526	0.9887	
	$t(20)$	0.9915	0.9810	0.9999		0.9865	0.9703	0.9998		0.7938	0.8914	0.9499	
	$t(30)$	0.9958	0.9765	0.9999		0.9847	0.9480	0.9995		0.7160	0.7971	0.8923	
	$t(40)$	0.8861	0.7902	0.9823		0.7516	0.5311	0.9156		0.2309	0.2944	0.1534	
	$t(50)$	0.9967	0.9962	1.0000		0.9893	0.9871	0.9996		0.6839	0.8868	0.8628	
	$t(5)$	0.8123	0.7485	0.9646		0.7834	0.7255	0.9563		0.8208	0.7866	0.9695	
	$t(10)$	0.7589	0.7382	0.9395		0.6955	0.7033	0.8978		0.5239	0.7090	0.7195	
$\Delta=1.50$	$t(15)$	0.8835	0.8028	0.9836		0.8888	0.8667	0.9836		0.7291	0.8122	0.9137	
	$t(20)$	0.9010	0.9116	0.9871		0.8690	0.9167	0.9779		0.6654	0.8528	0.8598	
	$t(30)$	0.8843	0.8721	0.9828		0.7995	0.8382	0.9471		0.5363	0.7131	0.6995	
	$t(40)$	0.3917	0.4297	0.4516		0.3620	0.4137	0.3959		0.2694	0.3532	0.2212	
	$t(50)$	0.6407	0.7543	0.8258		0.5580	0.7989	0.7245		0.3828	0.7148	0.4382	
	$t(5)$	0.8426	0.7485	0.9750		0.8194	0.7255	0.9683		0.8370	0.7866	0.9754	
	$t(10)$	0.7576	0.7485	0.9391		0.6814	0.7329	0.8855		0.5005	0.7010	0.5005	
	$t(15)$	0.9280	0.8348	0.9926		0.9349	0.8687	0.9945		0.7788	0.8373	0.9426	
	$t(20)$	0.9271	0.9386	0.9927		0.9025	0.9240	0.9876		0.6870	0.8750	0.8774	
	$t(30)$	0.9230	0.8752	0.9912		0.8547	0.8615	0.9721		0.5661	0.7308	0.7428	

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Declarations

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